

How the EdgeQuake demo handles real questions

A plain-English readout of a live test on the shared demo environment — what users can expect when they ask questions against an existing knowledge workspace.

ENVIRONMENT	TEST DATE
https://demo.edgequake.com · product v0.28.3	29 September 2026
TENANT	WORKSPACE
00000000-0000-0000-0000-000000000002	00000000-0000-0000-0000-000000000003
ANSWERING MODEL	EMBEDDING MODEL
Mistral · mistral-small-latest	OpenAI · text-embedding-3-small (1536-d)

Bottom line. On this demo tenant, knowledge questions were answered successfully in every sample we ran. A typical *new* question finishes in about **7–14 seconds**. If the same question is asked again, the system can answer in about **1–3 seconds** by reusing a cached answer. We did **not** find a breaking point (capacity “knee”) up to four people asking at once under a 25-second comfort target.

1. What we tested (and why)

EdgeQuake is a knowledge system: documents are turned into a searchable graph, then people ask questions in natural language. Because answers depend on large language models (LLMs), wall-clock “requests per second” alone can mislead. This snapshot follows SPEC-153: measure the **question-and-answer path** on a named tenant, separate **fresh answers** from **cached repeats**, and speak in outcomes a business owner can use.

We exercised the live demo API with authenticated calls scoped to the tenant and workspace above. Workload shape: interactive Q&A over the papers already loaded in the workspace (including text-to-SQL and agentic research PDFs).

2. Headline numbers

SUCCESS RATE	FRESH ANSWER TIME	ANSWER SPEED (FRESH)
100%	~7–14 s	~90 tok/s
All sampled questions returned a usable answer	Server time for a new question (typical / upper)	Average generation pace reported by the model path

SITUATION	QUESTIONS	TYPICAL TIME	UPPER (≈90%)	WHAT IT MEANS
Fresh questions (single user)	6	10.1 s	14.1 s	Realistic first-ask experience

SITUATION	QUESTIONS	TYPICAL TIME	UPPER (≈90%)	WHAT IT MEANS
New unique prompts (follow-up)	4	7.3 s	9.6 s	Confirms cold path without repeats
Repeated questions (cache)	14	~1–3 s	< 5 s	Same question answered from cache — very fast

Important nuance. When we raised concurrency to 2 and 4 users, nearly every response was served from the *answer cache* (the model did not regenerate). That is excellent for repeat FAQs, but it is **not** the same as proving how many brand-new LLM answers the system can produce at once. We therefore do not publish an inflated “800+ tokens per second capacity” figure from that window.

3. Where the seconds go (fresh answer)

On unique follow-up questions, average time inside the service broke down roughly as follows:

STAGE	PLAIN ENGLISH	AVG TIME	SHARE
Understand the question	Embed the query and extract search keywords	~2.8 s	
Find evidence	Search the graph and documents in the workspace	~0.9 s	
Write the answer	Language model generates the response	~4.2 s	

In business terms: more than half of a fresh answer is waiting on the language model. Retrieval in this workspace is comparatively quick. Improving “time to answer” for new questions is mostly an LLM / prompt-budget topic, not a “database is slow” topic — for this corpus size.

4. Capacity under light concurrent use

We gradually increased simultaneous questioners (1 → 2 → 4) while requiring answers within a **25-second** comfort target and a usable response every time.

Result: success stayed at 100%. We never hit a “breaking point” in this range. That supports a cautious statement: *the demo tenant comfortably handled at least four concurrent interactive asks under today’s conditions* — especially when some asks hit cache.

CONCURRENT USERS	SUCCESS	NOTES
1	100%	Fresh LLM answers
2	100%	Mostly cached repeats
4	100%	Mostly cached repeats

5. What this means for stakeholders

For product / customer success

- Demo Q&A is reliable on this tenant for the loaded research corpus.

- Set expectations: **first answer $\approx$ 10 seconds; repeat FAQ $\approx$ a couple of seconds.**
- This workspace uses **Mistral Small** for answers (not the fleet OpenAI default shown on the public health page).

For engineering / capacity planning

- Publish cold and warm numbers separately (SPEC-153 rule).
- A true saturation knee for *fresh* LLM answers still needs a longer cold-only sweep (and optionally a dedicated bench tenant).
- Ingest / PDF conversion capacity was intentionally **not** stressed on this shared demo.

Safe claims you can make today

- Demo knowledge Q&A worked in 100% of sampled asks.
- Fresh answers typically complete in about 7–14 seconds.
- Cached repeats are about 1–3 seconds.
- No failures observed up to four concurrent users under a 25-second target in this run.

Claims to avoid

- Do not cite “800+ tokens/s system capacity” from the concurrent window — that mixed cache hits.
- Do not claim a measured maximum capacity ceiling; we did not reach one.
- Do not equate fleet health “OpenAI” with this workspace’s answer model (Mistral).

6. How to read this next to other EdgeQuake reports

REPORT TYPE	QUESTION IT ANSWERS	THIS SNAPSHOT?
Accuracy (SPEC-001)	Are answers <i>correct</i> ?	No — we checked usefulness/success, not graded accuracy
Database physics (SPEC-090 / 063)	Is Postgres / search infrastructure healthy at scale?	No — corpus here is demo-sized
Workload / tokens (SPEC-153)	How does Q&A feel and scale for a tenant?	Yes

Evidence pack: [specs/153-workload-benchmark/measurements/2026-09-29-demo-query-knee/](#) · Protocol: SPEC-153 · Generated for business stakeholders from live measurements on the shared demo. Hosted environment caches could not be force-disabled; cold vs cache attribution uses server timing signals (`generation_time_ms = 0` $\Rightarrow$ answer cache).